Topic 5: Arms Race (Group 10)

Zeyn Jaswal, Azzaam Khan, Timothy Palin, Meredith Reeves, Tony Xu

Overview

Scenario: Purple (p) and Green (g) are two countries competing in an arms race.

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- How does public sentiment affect defence spending?
- How do limited resources affect defence spending?

Research Question: What happens to their arms spending over time?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.

Base Model

Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r \tag{1}$$

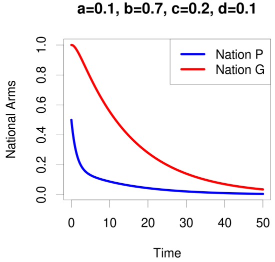
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s \tag{2}$$

- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c: Mutual Fear $(a, c > 0)$
- b, d: Spending Burden $(b, d > 0)$
- $(r, s) < 0$ for trust and $(r, s) > 0$ for hostility

Long Term Behavior ($r = s = 0$)

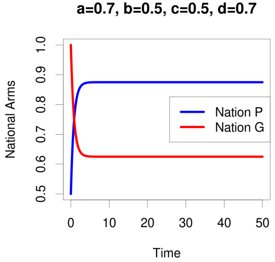
Assuming no underlying trust/hostility, we have:

$bd > ac$



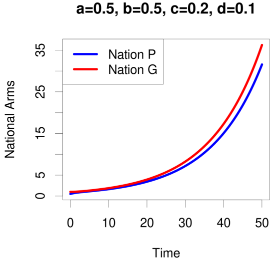
Complete mutual disarmament

$bd = ac$



Stabilized spending

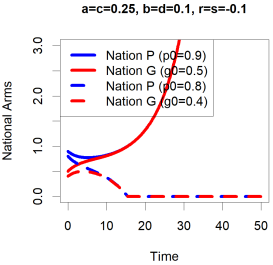
$bd < ac$



Perpetual arms race

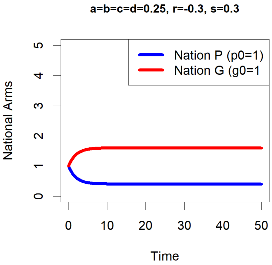
Long Term Behavior with Trust/Hostility

$bd < ac, r < 0, s < 0$



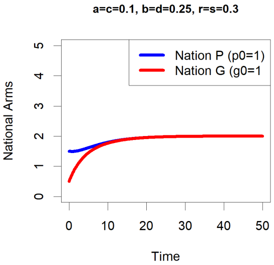
Arms equilibrium determined by initial spending

$bd = ac\ r < 0, s > 0$



Stable arms spending

$bd > ac, r > 0, s > 0$



Stable arms spending

Mutual Logistic Fear

Limitation of the Current Model: Each country has a limited amount of resources at a given time.

Solution: Introduce self-limiting logistic growth.

Model Equations

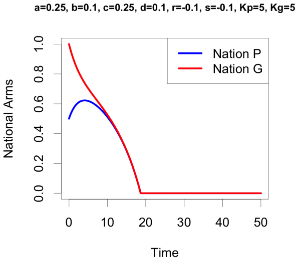
The equations after adding logistic growth are:

$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r \tag{3}$$

$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s \tag{4}$$

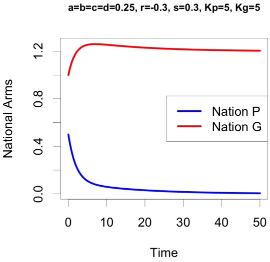
Long-Term Behavior of Mutual Logistic Fear

$bd < ac,\ r < 0,\ s < 0$



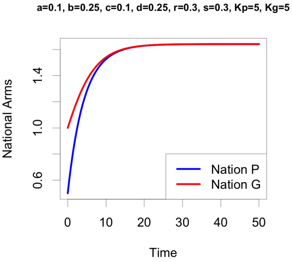
Mutual disarmament

$bd = ac,\ r < 0,\ s > 0$



Stabilized spending

$bd > ac,\ r > 0,\ s > 0$



Stabilized spending

Conclusion and Next Steps

Conclusion

- **Base Model:** In this model, long-term outcomes depend only on combined fear versus combined spending preservation.
- **Public Sentiment:** Initial spending influences this model's behavior and shifts the equilibrium which makes stable positive spending more likely.
- **Mutual Logistic Fear:** This model accounts for economic limitations and prevents unending arms race with infinite spending.

Next Steps

- Sensitivity Analysis
- Analytical solutions for equilibrium and stability

Thank you for listening!

[0] Presentation theme by Ricardo B. Lourenco